

Adaptive-Based Gamification System for Programming

"Smarter learning through adaptive intelligence."



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INSTITUTION	DEPARTMENT	CORE STACK	STUDY	ACADEMIC YEAR
Qassim University, Saudi Arabia	Computer Science	Python · Unity · Supabase	Mixed-Methods · 114 Participants	1446/1447 H — 2025/2026

1 INTRODUCTION

Gamification enhances learning by integrating game elements — challenges and progression — into educational environments.

This project presents an AI-driven gamified platform that teaches Python programming through:

- Interactive coding challenges
- Detective-style narrative progression
- Adaptive personalized learning paths

2 PROBLEM STATEMENT

Traditional programming education faces persistent challenges:

- Low engagement and motivation
- One-size-fits-all instruction
- No personalization for skill level
- Limited beginner progression support

Existing platforms rely on static gamification without adapting to learner needs — resulting in reduced long-term engagement.

3 PROPOSED SOLUTION

We developed an **Adaptive AI-based Gamified System** that:

- Dynamically adjusts difficulty based on performance
- Integrates Python tasks into a narrative storyline
- Provides personalized learning paths
- Supports gradual, measurable skill progression

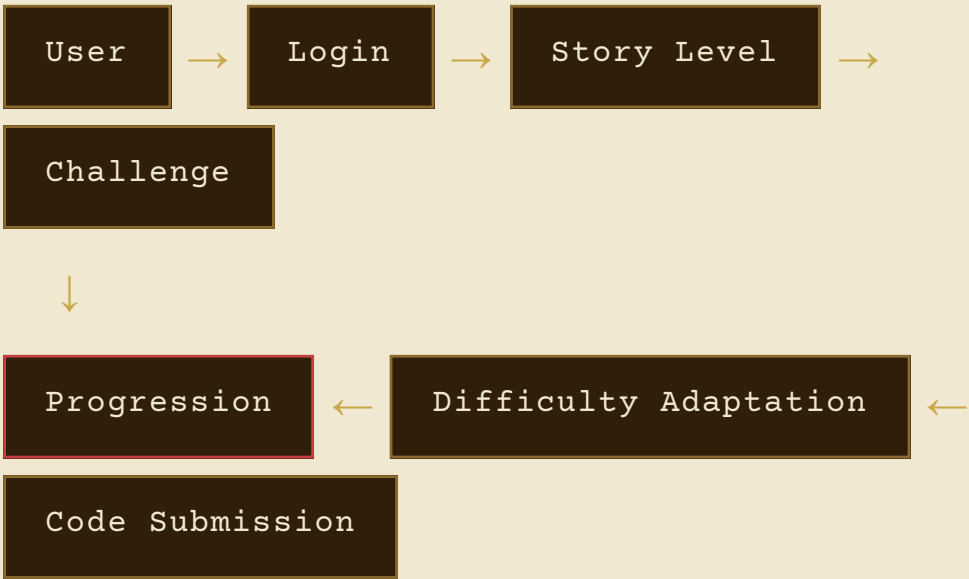
ADAPTIVE LEARNING

STORYTELLING

PROBLEM SOLVING

4 SYSTEM DESIGN

MODEL FLOW



5 METHODOLOGY

RESEARCH APPROACH

- Mixed-methods: Quantitative + Qualitative
- Iterative prototyping and evaluation

114

SURVEY PARTICIPANTS

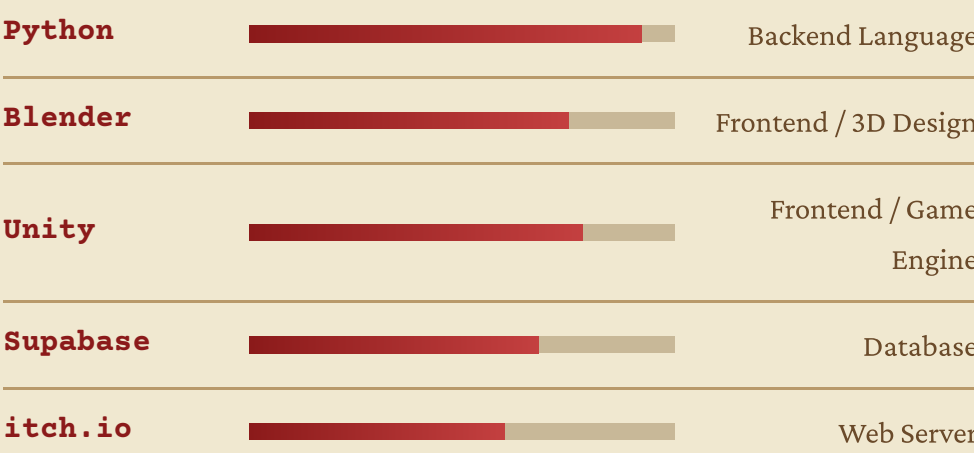
KEY FINDINGS

- Majority: beginner to intermediate learners
- Gamification significantly increases motivation
- Users prefer level-aware, adaptive systems

Confirms the necessity of adaptive AI-driven learning in programming education.

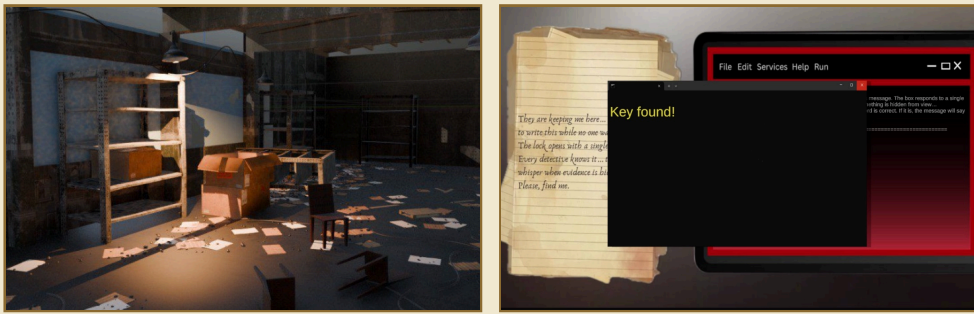
6 TOOLS & TECHNOLOGIES

TECHNOLOGY STACK



7 RESULTS & OUTPUT

The system was tested with 10 users starting at Level 1 (Easy). Across the gameplay sessions, the adaptive difficulty engine responded dynamically to individual performance — participants who solved challenges quickly were advanced to harder levels, while those who struggled received more support. This confirmed the engine functions as intended, personalizing each learner's path rather than applying a fixed progression.



ADAPTIVE DIFFICULTY FUNCTIONS AS INTENDED ✓

8 CONCLUSION

This system demonstrates how combining gamification with adaptive AI can address the core weaknesses of traditional programming education. By embedding Python learning within a detective-style narrative and adjusting challenge difficulty in real time, the platform achieves meaningful educational outcomes:

- Personalizes the learning experience per learner
- Increases engagement and motivation over time
- Improves programming education outcomes measurably
- Supports learners at beginner and intermediate levels

The integration of narrative-based design with AI-driven adaptation represents a significant step forward in intelligent educational systems for programming languages.



PROJECT REPOSITORY

Graduation Project
Dept. of Computer Science
College of Computer
Qassim University, Saudi Arabia
1446/1447 H (2025/2026)